Raspberry Pi Pico

P1
rpi pico 1 W
2648-SC0918CT-ND
smd

+3.3V
R1 220
3V1 green
GND

R10 4.7k
GND

+BATT
+3.3V
GND
5V
J1

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

D1 D_Schottky
5V
+3.3V

R2 220
D13 blue
GND

bt1 bt2 bt3 bt4
sda
scl
right_2 right_1 right_en
left_2 left_en left_1

J3 joystick
J4 speed knob

3V3_EN
RUN
VBUS VSYS 3V3_OUT ADC_VREF
GP0 GP1 GP2 GP3 GP4 GP5 GP6 GP7 GP8 GP9 GP10 GP11 GP12
GP13 GP14 GP15 GP16 GP17 GP18 GP19 GP20 GP21 GP22 GP26 GP27 GP28
AGND GND GND_1 GND_2 GND_3 GND_4 GND_5 GND_6

* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.

left motor

U1
DRV8245HQPWPRQ1

LVL6, in independent mode that means faults are retried

SR DIAG PH/IN2 EN/IN1 DRVOFF VM VM VM OUT1 OUT1 OUT1 GND GND GND
EPAD ITRIP MODE NFAULT IPROPI NSLEEP VM VM VM OUT2 OUT2 OUT2 GND GND GND
EPAD GND GND
8.2k +-10% LVL2 INDEPENDENT
8.2k R7 GND
left_ipropi
left_en
left_2
left_1
+BATT
GND
C1 0.1uF
HSL1 Heatsink

J100 0197054001
J101 0197054001
optionally replace J7 to be weelye compatible
ML+ ML-
J7 1017491

right motor

U2
DRV8245HQPWPRQ1

SR DIAG PH/IN2 EN/IN1 DRVOFF VM VM VM OUT1 OUT1 OUT1 GND GND GND
EPAD ITRIP MODE NFAULT IPROPI NSLEEP VM VM VM OUT2 OUT2 OUT2 GND GND GND
EPAD GND GND
8.2k +-10% LVL2 INDEPENDENT
8.2k R6 GND
right_ipropi
right_en
right_1
right_2
+BATT
GND
C2 0.1uF
HSR1 Heatsink

MR- MR+
J8 1017491
optionally replace J8 to make the PCB weelye compatible
wellye style J104
wellye style J105
J106 0197054001
J107 0197054001

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR
+3.3V NT1 NetTie_3
TP0 TestPoint
ADDR AIN0 AIN1 AIN2 AIN3 NC VDD SCL SDA GND
scl sda
right_ipropi
left_ipropi
+BATT
R13 100k
R14 8.2k
C4 0.1uF
pullup resistors
TP1 TestPoint left_ipropi
TP2 TestPoint right_ipropi
R11 604
R12 604
R15 3.3k
R16 3.3k

qwiic I2C plug
GND +3.3V sda scl
J2 Conn_01x04

weelye compatible header
J11 Conn_01x10_Pin
+BATT
switch
bat-
D2 D_Schottky

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1 B2 B3 B4
ASJ-200-B-HT
bt1 bt2 bt3 bt4
GND

battery power circuit
J102 0197054001
J103 0197054001
optionally replace J5 to be weelye compatible
J5 TB005-762-02BE
screw terminal
SW1
switch
R3 3.3k
R4 22.1k
R5 22.1k
18V zener
D4 D3
PSMN1R0-40YLDX PSMN1R0-40YLDX
Q1 on/off switch
Q2 reverse polarity protection
C5 33uf
J6
bat-
+BATT
GND

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

alternative voltage regulator

Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.

U5
D78B05T, RBT05W24S05, pololu5593
+BATT IN GND OUT 5V
GND

optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6V)
M1 M2

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=FA36C6A68A48DC1D

U3
LMR54410DBVR
CB GND FB SW VIN EN
+BATT
C6 0.1uF
C3 1uF
C7 0.1uF
L1 10uH
R8 115k
R9 22.1k
C8 22uF
5V
GND

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title: GBG-PCB
Size: A3 | Date: | Rev: v2.3
KiCad E.D.A. 10.0.6 | Id: 1/1